

• EXECUTIVE BRIEFING • 2026

Global Health Intelligence

An event-based surveillance platform that watches the world for disease events, judges them against international law, and puts the ones that matter in front of a Saudi analyst — in hours, not weeks.

59

sources monitored

114

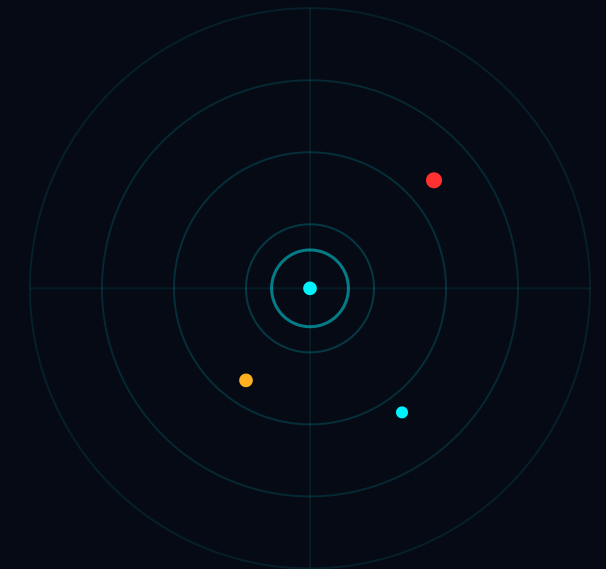
countries reporting

24/7

automated collection

~350

SAR per month, all-in



THE PROBLEM

Outbreaks are reported before they are confirmed

The first signal of an emerging event is almost never a laboratory result. It is a news report, a ministry statement, or a physician noticing something unusual — arriving scattered across dozens of agencies, in several languages, at all hours.

BEFORE

Manual scanning

An analyst opens the same thirty websites each morning. Coverage depends on who is on shift, how much time they have, and which languages they read.

THE GAP

Volume beats attention

There is no shortage of health news. There is a shortage of judgement about which item, out of hundreds, will matter to the Kingdom next month.

WHAT GHI DOES

Collect, score, escalate

Machines do the reading and the arithmetic. People do the judgement. Everything the system decides is recorded with the reasoning that produced it.

This is the model WHO calls **Event-Based Surveillance** — the same approach behind WHO's own EIOS platform. GHI applies it with Saudi priorities built into the scoring.

• HOW IT WORKS

Two streams in, one decision out

Every item follows the same path and no step is skipped. Verification comes before risk assessment, exactly as WHO's event-based surveillance requires — a rumour cannot escalate itself.



AUTOMATED

Steps 3 and 5 run without a person. **Every automated output is a draft.** The analyst's answer is stored separately and always takes precedence — the machine's original position stays on the record, so any disagreement is visible and auditable.

• THE METHOD

Scored against international law, not intuition

Priority is not a feeling. Every signal is scored on the four questions the International Health Regulations (2005) Annex 2 requires a State Party to ask.

QUESTION 1

Serious impact?

Judged against the expected level for that disease in that country — not against raw case counts.

QUESTION 2

Unusual or unexpected?

Novelty is the strongest single predictor of a public health emergency of international concern.

QUESTION 3

Risk of spread?

Healthcare-worker infection weighs heavily — the classic sentinel for sustained human transmission.

QUESTION 4

Travel or trade risk?

Border measures and advisories carry direct economic consequence for the Kingdom.

THE ESCALATION RULE

Any two of four means notify WHO.

That is Annex 2's own threshold, not a number this system invented. A high score is therefore a *prediction of the legal outcome*, not a separate opinion about it.

ANOMALY, NOT MAGNITUDE

Forty cholera cases in Yemen during an ongoing epidemic is background. **Four in Riyadh is an emergency.**

45 disease baselines encode what “expected” means per disease and country, so severity is measured as deviation.

• COVERAGE

59 sources, ranked by how much they can be believed

Not all reporting is equal. Each source carries a credibility score, and confidence rises when independent sources agree — the strongest evidence this kind of surveillance produces.

TIER	EXAMPLES	SCORE
Multilateral & national authorities	WHO, US CDC, ECDC, Africa CDC, PAHO, UKHSA	100
National ministries of health	Saudi MOH, RKI, Japan MHLW, China CDC	95
Early-warning networks	ProMED, Global.health	90
News agencies	Reuters, AP, AFP	85
Humanitarian reporting	ReliefWeb, OCHA, IFRC	80
Specialist health press	CIDRAP	75
Preprint servers	medRxiv, bioRxiv	65
Social media	X — always requires corroboration	30

WHY PREPRINTS SIT BELOW REUTERS

A preprint has not been peer reviewed. Treating it as “scientific, therefore reliable” is how a paper that is later withdrawn becomes a national alert.

COVERAGE EXTENDS BEYOND HUMAN DISEASE

- **One Health** — animal disease sources, because most emerging pathogens are zoonotic
- **Environmental & CBRN** — chemical and radiological events, not only biological
- **GCC & neighbours** — weighted for borders, corridors and pilgrim flows
- **Arabic-language** reporting across the region

ANALYST EXPERIENCE

The assessment is written before the analyst opens it

Accepting a signal produces a completed IHR Annex 2 answer set and Rapid Risk Assessment, with the reasoning behind each answer. The analyst edits rather than composes.

EXAMPLE — GENERATED, UNEDITED

CRITICAL

NOTIFY WHO

CONFIDENCE: HIGH

MERS-CoV — Saudi Arabia

WHO EMRO · 2 of 4 Annex 2 questions answered yes

Q1 · Serious public health impact

YES

Q2 · Unusual or unexpected

NO

Q3 · Risk of international spread

YES

Q4 · Travel or trade measures

NO

“Observed CFR 39.0% is at or above the 35% baseline. Respiratory transmission route. MERS-CoV is endemic in Saudi Arabia; occurrence is expected rather than novel.”

THE ANALYST ALWAYS WINS

The draft is frozen the moment it is written. Analyst edits are stored separately, so the difference between the two **is** the override record — no precedence rule, nothing to configure, and the machine's original position remains available for audit.

NO MODEL DECIDES RISK

Language models read facts out of source pages — disease, place, counts, dates. **Every risk classification is deterministic code.** A health authority must be able to explain why something was rated Critical, and “the model judged it so” is not an explanation that survives review.

UNCERTAINTY IS STATED, NOT HIDDEN

Each draft lists what is *missing* — no case counts, no baseline on record, a single uncorroborated source. Those are precisely the things overlooked under load.

• OPERATING TODAY

What the platform is doing right now

Live production figures, not projections. The system runs unattended on a six-hourly cycle. 249 events have been collected in total; the figures below are those currently actionable — informational items are retained but not surfaced.

184

actionable events in the current window

114

countries reporting

19

cleared the IHR threshold

16

promoted to triage automatically

32

sources collecting cleanly

51

social signals from X

64

watch terms — 28 in Arabic

45

disease baselines encoded

CURRENTLY IN THE ANALYST QUEUE

CRITICAL

MERS-CoV — Saudi Arabia

CRITICAL

Meningococcal disease — Sudan

CRITICAL

Wild poliovirus — Pakistan

HIGH

Cholera — Yemen

HIGH

Avian influenza H5N1 — Egypt

HIGH

Measles — Uganda

• ARCHITECTURE

Serverless, with one small machine for the hard pages

Nothing to patch on a Friday night. The platform scales to zero when idle and costs nothing while nothing is happening.

EDGE

Cloudflare

- **Workers** — API and scheduled collector
- **Pages** — the analyst interface
- **Hyperdrive** — pooled database access

DATA

Neon Postgres

- Serverless Postgres, London region
- Every score stored with its evidence
- Point-in-time recovery

RENDERING

Dedicated crawler

- Hetzner, Germany — 2 vCPU / 4 GB
- Runs a real browser for pages that need one
- Unlocks RKI, China CDC, WHO dashboards

SECURITY POSTURE

- Every endpoint requires a session — **nothing is public**
- Access restricted to **@pha.gov.sa** accounts
- Role-based permissions per view
- Only open-source reporting is collected — no personal data

COST CONTROL BY DESIGN

- Pages re-read only when their content changes
- Feeds send only entries not already seen
- Per-source spend recorded and reportable
- Hard budget caps on every paid interface

TOTAL COST OF OWNERSHIP

Around 350 riyals a month

Measured from live operation, not estimated. No licences, no per-seat charges, no minimum commitments.

COMPONENT	PURPOSE	SAR / MONTH
Hetzner CX22	Browser rendering	120
X (Twitter) API	Social listening	100
Claude API	Reading source pages	~75
Neon Postgres	Database	35
Cloudflare Workers	API, interface, scheduling	20
Total		≈ 350

The Claude API line is usage-based and varies with how much the world's health agencies publish. A worst-case month — every source changing on every scan — would reach roughly 300 SAR; measured operation runs well below that.

IN CONTEXT

4,200 SAR

per year, fully operational — roughly the cost of one staff-day per month.

WHAT ENGINEERING CHANGED

Original design ~3,750 SAR / mo

Today ~350 SAR / mo

Achieved by reading each page only when it changes, sending only unseen feed entries, and matching the model to the task rather than using the largest available for everything.

• WHERE IT GOES NEXT

Operational today, stronger each quarter

The platform is running in production. The work ahead deepens judgement rather than adding surface.

DELIVERED

- ✓ 59-source registry with credibility tiers
- ✓ IHR Annex 2 scoring and auto-promotion
- ✓ Machine-drafted assessments with human override
- ✓ Social listening in Arabic and English
- ✓ Browser rendering for JavaScript-heavy sources
- ✓ Full audit trail on every decision

NEXT

- ▶ **Disease name normalisation** — so two agencies reporting one outbreak count as corroboration
- ▶ **PDF extraction** — several agencies publish only as attachments
- ▶ **Hajj & Umrah windows** — raised sensitivity during mass gatherings
- ▶ **Baseline curation** by PHA epidemiologists

UNDER CONSIDERATION

- ▶ Scheduled executive briefing by email
- ▶ Arabic interface for the analyst views
- ▶ Integration with national notification workflows
- ▶ Historical trend analysis across seasons

THE MOST VALUABLE DATASET IS NOT CODE

The 45 disease baselines — what counts as expected, per disease, per country — are the highest-leverage asset in the system. They currently hold indicative starting values. **Curation by PHA epidemiologists would improve every score the platform produces.**

• SUMMARY

Earlier warning, defensible judgement, at negligible cost

59

sources watched continuously, in two languages, without a person opening a browser

4

IHR Annex 2 questions answered on every signal, with the evidence recorded

350

riyals per month to operate the entire platform

GHI does not replace an epidemiologist. It ensures that when one sits down, the work in front of them is the work that matters — and that every decision the Authority takes can be explained, months later, to whoever asks.